22.4. INJECTION OF TRU INTO THE ENVIRONMENT

Of the artificial radionuclides released to the environment by nuclear activities, the transuranium (TRU) species are a major concern. This concern arises from the very long half-life of a number of the nuclides as well as their high radiotoxicity values. Although reactor operation and spent fuel reprocessing activities have released small amounts of TRU's to the environment, testing of nuclear weapons has released rather large quantities. Since the first nuclear test detonation in New Mexico in 1945, approximately 3 500 kg of plutonium has been released in atmospheric explosions and another 100 kg in underground tests. This corresponds to about 11 PBq of $^{239+240}$Pu ejected into the atmosphere. In addition 0.6 PBq of ^{238}Pu were released over the south Pacific in the high altitude destruction of the SNAP-9 satellite power source in 1964. By contrast, a total of ~0.58 PBq of $^{239+240}$Pu has been released into the Irish Sea from the Sellafield (UK) reprocessing plants between 1971 and 1999; most of this before 1985. About 37 kg of ^{241}Am is present in the environment from the decay of ^{241}Pu from the nuclear testing. Table 22.6 compares the TRU's released in nuclear tests in the atmosphere, in the Irish Sea from the Sellafield plants, and into the English Channel from the LaHague plants. As the amount of spent nuclear fuel increases, the contribution to the total plutonium in the environment could become more significant over a longer time, especially if nuclear waste disposal sites release actinide elements slowly to the environment. Whatever the sources of plutonium and other actinides, their presence represents a contamination of the environment by highly toxic material. An understanding of the factors involved in their retention and/or migration in the ecosphere is therefore highly desirable. Studies of the environmental behavior of releases from tests provide data needed to understand and predict the behavior of smaller releases from the nuclear power industry. Therefore, in the next few paragraphs we concentrate our discussion to the behavior of the actinide elements in the environment, particulary the mobility of plutonium.

Table 22.6 TRU released from nuclear tests and from reprocessing (UNSCEAR2000, BNFL and COGEMA)

	Tests (TBq)	Sellafield (TBq)		LaHague (TBq)	
Nuclide	**1945–1980**	**1971–84**	**1985–94**	**1995–99**	**2000**
^{237}Np	–	–	0	0.33	
^{239}Pu	6520				
^{240}Pu	4350	559	15	0.92	
^{241}Pu	142000	–	–	21.8	0.039[†]
^{241}Am	–	442	9	0.31	
^{242}Cm	–	–	–	0.052	
$^{243+244}$Cm	–	–	–	0.024	

[†]Total α.

The majority of the plutonium from weapons testing was injected initially into the stratosphere. The plutonium originally in the weapon which survived the explosion would have been formed into high-fired oxide which would be expected to remain insoluble as it returned to earth. Such insoluble particles would have sunk in a rather short time into the bottom sediments of lakes, rivers, and oceans or would become incorporated in soils below the surface layer. However, in most nuclear weapon explosions a considerable amount of plutonium is generated in the explosion via ^{238}U (n,γ) reactions and subsequent α-decay of the product ^{239}U, ^{240}U, ^{241}U, etc. In total, about two thirds of the plutonium released was generated in this way. The nuclides from the (n,γ) reactions would exist as single atoms, and, hence, were never formed into high-fired oxides. The plutonium from this formation path would have been soluble and, as a result, more reactive and its behavior would be more similar to that of plutonium released from nuclear reactors, reprocessing plants and from nuclear waste repository sites.

22.5. PRESENT LEVELS OF TRU IN THE ECOSPHERE

The United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR), in 2000 reports data that corresponds to the following total average global nuclide depositions from atmospheric nuclear weapon tests: ^{89}Sr 11, ^{90}Sr 1.2, ^{95}Zr 19, ^{106}Ru 12, ^{137}Cs 1.8, ^{239}Pu 0.013, ^{240}Pu 0.009, and ^{241}Pu 0.278 kBq m^{-2}. Some of these radionuclides have now decayed because of their short half-lives and the end of atmospheric testing in 1980 and almost all of the ^{241}Pu has decayed to ^{241}Am.

Near test sites, reprocessing facilities, etc., the concentration of plutonium in the soil and water is much higher than in more distant locations. Generally, the great majority of plutonium is associated with sub-surface soils or sediments or with suspended particulates in water. For example, when vegetation, animals, litter and soils are compared, $\geq$99% of the plutonium is present in the soil. Similarly, in shallow bodies of water, more than 96% of the plutonium is found associated with the sediments. However, it is via the species that are soluble or attached to suspended colloids and/or particulate matter in water that plutonium is transported in the environment. Analysis of vertical plutonium migration in soils near Chernobyl and in eastern Europe from the Chernobyl accident has shown that most of the plutonium is still in the first 0.5 cm from the surface for soils with significant humic acid content. In these soils, the plutonium is mostly associated with the insoluble calcium-humate fraction. In non-humic, carbonate rich soils, the plutonium has moved several centimeters downward. Migration rates of $\leq$0.1 cm y^{-1} is associated with the humic soils and of 1–10 cm y^{-1} with the carbonate rich ones. Presumably, migration is retarded by the interaction with the immobilized humic material in soils.